The fast linear operator of the Euler equations in potential-temperature form: hybridized DG with LMARS flux, relaxed trace conditions and time integration without elimination

Documentation of `hdg_relax_theta.py` / `HDGRelaxTheta.jl`

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Abstract

We derive and discretize the linear fast operator of the one-dimensional (vertical) compressible Euler equations in the variables $(\rho, \rho w, \rho\theta)$, as needed for the fast (acoustic) part of a split-explicit method. Momentum advection is removed, θ is frozen in the $\rho\theta$ equation, and the pressure is written as $p = \frac{p}{\rho\theta}\,\rho\theta$ with $\frac{p}{\rho\theta}$ frozen. The resulting system is linear in the full state. It is discretized by a hybridized DG (HDG) method whose stabilization $\boldsymbol{T} = |\boldsymbol{A}|$ reproduces the LMARS flux exactly. The trace conditions are relaxed into evolution equations, and the coupled element/trace system is integrated without static condensation, either monolithically (SDIRK4) or with an IMEX scheme whose implicit stage is local per face. Frozen coefficients are continuous piecewise polynomials, which makes discrete hydrostatic states exactly steady. The operator accepts a constant slow tendency, so it can be used as the fast part of a split-explicit scheme.

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1 Model and linearization

1.1 Full equations

In a vertical column with $m = \rho w$ and $\Theta = \rho\theta$,

$$\partial_t \rho + \partial_z m = 0, \tag{1}$$

$$\partial_t m + \partial_z (mw + p) = -\rho g, \tag{2}$$

$$\partial_t \Theta + \partial_z (\theta m) = 0, \qquad p = p_{00}\Big(\frac{R\,\Theta}{p_{00}}\Big)^{\gamma}. \tag{3}$$

1.2 Linearization (fast operator)

Let a frozen state (bars) be given, for example the current slow state or a reference profile. The linearization consists of three steps.

1. Remove momentum advection: $\partial_z(mw) \to 0$.
2. Freeze θ in the Θ equation: $\partial_z(\theta m) \to \partial_z(\bar\theta\, m)$.
3. Write $p = \dfrac{p}{\Theta}\,\Theta$ and freeze the ratio: $p \to \bar\pi\,\Theta$, with $\bar\pi = \bar p/\bar\Theta = R\,\bar\Pi$ (Π = Exner function).

This gives

$$\boxed{\partial_t \rho + \partial_z m = 0, \qquad \partial_t m + \partial_z (a\,\Theta + b) = -g\rho, \qquad \partial_t \Theta + \partial_z(\bar\theta\, m) = 0,} \tag{4}$$

with $a = \bar\pi$ and $b = 0$. The system is linear in the *full* state $\boldsymbol{q} = (\rho, m, \Theta)$, not in perturbations. In flux form, $\partial_t \boldsymbol{q} + \partial_z(\boldsymbol{A}\boldsymbol{q} + b\,\boldsymbol{e}_m) = \boldsymbol{S}\boldsymbol{q}$ with

$$\boldsymbol{A}(z) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & a \\ 0 & \bar\theta & 0 \end{pmatrix}, \qquad \boldsymbol{S} = \begin{pmatrix} 0 & 0 & 0 \\ -g & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{5}$$

1.3 Sound speed and the factor `pfac`

The eigenvalues of $\boldsymbol{A}$ are $0, \pm c$ with

$$c^2 = a\,\bar\theta = \frac{\bar p}{\bar\Theta}\,\frac{\bar\Theta}{\bar\rho} = \frac{\bar p}{\bar\rho} = R\bar T. \tag{6}$$

Freezing p/Θ therefore gives the isothermal (Newtonian) sound speed, smaller than the adiabatic one, $\gamma R\bar T$, by a factor $\sqrt{\gamma}$ (287 versus 340 m/s at 288 K). The tangent linearization would use $\partial p/\partial\Theta = \gamma p/\Theta$. In a split-explicit scheme the slow tendency can carry the remainder $p - \bar\pi\Theta$, so the choice is a design decision. The code provides

$$p_{\text{lin}} = a\,\Theta + b, \qquad a = \texttt{pfac}\,\bar\pi, \qquad b = (1 - \texttt{pfac})\,\bar p, \tag{7}$$

where `pfac` $= 1$ (default) is the linearization above and `pfac` $= \gamma$ is the tangent-linear pressure with $c^2 = \gamma R\bar T$. The constant b keeps $p_{\text{lin}}(\bar\Theta) = \bar p$, so the frozen hydrostatic state remains steady for any `pfac`.

1.4 Use in a split-explicit method

With the full nonlinear operator $\mathcal{N}$ and the fast operator $\mathcal{L}$ of (4), frozen at the slow level $\boldsymbol{q}^n$, the fast system integrated over the small steps is

$$\partial_t \boldsymbol{q} = \mathcal{L}(\boldsymbol{q}) + \boldsymbol{F}_{\text{slow}}, \qquad \boldsymbol{F}_{\text{slow}} = \mathcal{N}(\boldsymbol{q}^n) - \mathcal{L}(\boldsymbol{q}^n) \ \text{ (or a multistage variant)}, \tag{8}$$

with $\boldsymbol{F}_{\text{slow}}$ constant over the small steps (e.g. [5]). Every run accepts `F_slow` as a constant tendency. The function `tendency(q)` returns $\mathcal{L}(\boldsymbol{q})$ in the HDG limit, so $\boldsymbol{F}_{\text{slow}}$ can be formed directly.

2 HDG discretization with LMARS flux

2.1 Local problem

On $K_e = [z_{e-1/2}, z_{e+1/2}]$ the element unknowns lie in $P_p(K_e)^3$ (Legendre basis), and each face carries a single-valued trace $\hat{\boldsymbol{q}}_f \in \mathbb{R}^3$. For all $\boldsymbol{v} \in P_p(K)^3$,

$$(\boldsymbol{v}, \partial_t \boldsymbol{q}_h)_K = (\partial_z \boldsymbol{v}, \boldsymbol{A}\boldsymbol{q}_h + b\,\boldsymbol{e}_m)_K + (\boldsymbol{v}, \boldsymbol{S}\boldsymbol{q}_h)_K - \sum_{f \in \partial K} \boldsymbol{v}(z_f)^{\mathsf{T}} \widehat{\boldsymbol{F}}_{K,f}, \tag{9}$$

$$\widehat{\boldsymbol{F}}_{K,f} = \big(\boldsymbol{A}_f \hat{\boldsymbol{q}}_f + b_f \boldsymbol{e}_m\big)\, n + \boldsymbol{T}_f\big(\boldsymbol{q}_h|_K(z_f) - \hat{\boldsymbol{q}}_f\big). \tag{10}$$

2.2 Stabilization and the LMARS flux

Since the eigenvalues of $\boldsymbol{A}$ are $0, \pm c$, we have $|\boldsymbol{A}| = \boldsymbol{A}^2/c$, i.e.

$$\boldsymbol{T}_f = |\boldsymbol{A}_f| = \begin{pmatrix} 0 & 0 & a_f/c_f \\ 0 & c_f & 0 \\ 0 & 0 & c_f \end{pmatrix}, \qquad c_f^2 = a_f \bar{\theta}_f. \tag{11}$$

Trace conditions.

- Interior faces: conservativity $\boldsymbol{T}_f(\boldsymbol{q}_L - \hat{\boldsymbol{q}}_f) + \boldsymbol{T}_f(\boldsymbol{q}_R - \hat{\boldsymbol{q}}_f) = \boldsymbol{0}$. The m and Θ rows give $\hat{m} = \{\{m\}\}$ and $\hat{\Theta} = \{\{\Theta\}\}$. $\hat{\rho}$ enters no flux, because the first columns of $\boldsymbol{A}$ and $\boldsymbol{T}$ vanish, and is set to $\{\{\rho\}\}$. So $\hat{\boldsymbol{q}} = \{\{\boldsymbol{q}\}\}$.
- Walls: $\hat{\boldsymbol{q}}_f = \mathrm{diag}(1, 0, 1)\, \boldsymbol{q}_{\mathrm{int}}$, i.e. $\hat{m} = 0$.

Inserting $\hat{\boldsymbol{q}} = \{\{\boldsymbol{q}\}\}$ into (10) with $[\![\cdot]\!] = (\cdot)_R - (\cdot)_L$ gives

$$\widehat{\boldsymbol{F}} = \begin{pmatrix} m^* \\ p^* \\ \bar{\theta}_f\, m^* \end{pmatrix}, \qquad m^* = \{\{m\}\} - \frac{a_f [\![\Theta]\!]}{2c_f} = \{\{m\}\} - \frac{[\![p_{\mathrm{lin}}]\!]}{2c_f}, \qquad p^* = a_f\{\{\Theta\}\} + b_f - \frac{c_f}{2}[\![m]\!]. \tag{12}$$

This is the LMARS flux [1] for (4): interface momentum and pressure from the acoustic Riemann invariants, and the Θ flux $\bar{\theta}\, m^*$, in which upwinding of θ is irrelevant because θ is frozen. The code checks this identity to round-off.

2.3 Frozen coefficients

The coefficients a, $\bar{\theta}$, b are represented as *continuous* piecewise polynomials of degree p, interpolated at Gauss–Lobatto nodes. This matches the situation in a DG model, where they come from the slow state (one would take a continuous reconstruction there as well). All volume integrands are then polynomials of degree at most $3p$. The code uses $2p + 2$ Gauss points, which integrate them exactly, and face values are single-valued.

2.4 Matrix form and relaxation

With all element coefficients in $\boldsymbol{q}$ and traces in $\hat{\boldsymbol{q}}$,

$$\begin{pmatrix} \boldsymbol{M} & \boldsymbol{0} \\ \boldsymbol{0} & \varepsilon \boldsymbol{I} \end{pmatrix} \frac{d}{dt} \begin{pmatrix} \boldsymbol{q} \\ \hat{\boldsymbol{q}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{K} & \boldsymbol{G} \\ \boldsymbol{H} & -\boldsymbol{I} \end{pmatrix} \begin{pmatrix} \boldsymbol{q} \\ \hat{\boldsymbol{q}} \end{pmatrix} + \begin{pmatrix} \boldsymbol{f}_b + \boldsymbol{M}\boldsymbol{F}_{\mathrm{slow}} \\ \boldsymbol{0} \end{pmatrix}. \tag{13}$$

Here $\boldsymbol{f}_b$ collects the affine pressure part b, and $\boldsymbol{H}$ is the face-local trace condition ($\hat{\boldsymbol{q}} = \boldsymbol{H}\boldsymbol{q}$). For $\varepsilon = 0$ this is the HDG/LMARS scheme as an index-1 DAE. For $\varepsilon > 0$ the trace condition is relaxed. In both cases the element unknowns are *not* eliminated. The relaxation error is $\mathcal{O}(\varepsilon c/h)$, because the trace defect $\hat{\boldsymbol{q}} - \boldsymbol{H}\boldsymbol{q} \approx -\varepsilon \boldsymbol{H}\dot{\boldsymbol{q}}$ enters through $\boldsymbol{M}^{-1}\boldsymbol{G} \sim c/h$.

3 Time integration

- `method="dirk"`: five-stage, fourth-order, L-stable and stiffly accurate SDIRK of Hairer–Wanner [4] ($a_{ii} = 1/4$). Each stage solves $(\mathcal{M} - \frac{\Delta t}{4}\boldsymbol{L})\boldsymbol{Y}_i = \mathcal{M}\boldsymbol{U}^n + \Delta t \sum_{j<i} a_{ij}(\boldsymbol{L}\boldsymbol{Y}_j + \boldsymbol{f}) + \frac{\Delta t}{4}\boldsymbol{f}$ with one sparse LU of the full coupled matrix. It is valid for every $\varepsilon \geq 0$.
- `method="imex"`: ARS(2,2,2) [3], with the element part explicit and the relaxation implicit. The implicit stage is $\hat{\boldsymbol{q}}^{(i)} = (\hat{\boldsymbol{q}}^* + k\boldsymbol{H}\boldsymbol{q}^{(i)})/(1+k)$ with $k = \Delta t\, a^I_{ii}/\varepsilon$. It is local per face, so there is no linear solve. It is asymptotic preserving: for $\varepsilon \to 0$ it becomes an explicit DG/LMARS scheme under the CFL condition $\Delta t \lesssim h/(c(2p+1))$.

The explicit variant is the natural fast integrator of a split-explicit method. The monolithic variant gives a vertically implicit (HEVI-type) option.

4 Properties

Proposition 1 (Discrete hydrostatic balance)**.** *Let Θ_h be continuous and piecewise P_p, $m_h = 0$, and $\rho_h = -\Pi_p\, \partial_z(a\Theta_h + b)/g$ element-wise ($\Pi_p = L^2$ projection), with $\hat{\boldsymbol{q}} = \boldsymbol{H}\boldsymbol{q}_h$. Then $(\boldsymbol{q}_h, \hat{\boldsymbol{q}})$ is a steady state of (13) with $\boldsymbol{F}_{\text{slow}} = 0$, for every $\varepsilon \geq 0$ and every `pfac`.*

Proof. Θ_h is continuous and $m_h = 0$, so $\hat{\Theta}_f = \Theta_h(z_f)$, $\hat{m}_f = 0$ and $\boldsymbol{H}\boldsymbol{q} - \hat{\boldsymbol{q}} = 0$. In (10), the ρ row $\hat{m}\, n + \frac{a}{c}(\Theta_h - \hat{\Theta})$ and the Θ row $\bar{\theta}\hat{m}\, n + c(\Theta_h - \hat{\Theta})$ vanish, and so do their volume terms. The m row of the flux is $p_{\text{lin}}(z_f)\, n$, with $p_{\text{lin}} = a_h\Theta_h + b_h$ continuous. Since the quadrature is exact, $(\partial_z v, p_{\text{lin}})_K - [v\, p_{\text{lin}}]_{\partial K} - g(v, \rho_h)_K = -(v, \partial_z p_{\text{lin}} + g\rho_h)_K = 0$ for all $v \in P_p$, by the definition of ρ_h. □

The code builds such states with `balanced_state` (Julia) or `hydrostatic_rho` (Python). An analytic background that is merely projected is steady only up to truncation error, $\mathcal{O}(h^p)$ in $\partial_t \boldsymbol{q}$ (Table 3).

Conservation. Summing the two one-sided fluxes at a face gives $\boldsymbol{T}_f(\boldsymbol{q}_L + \boldsymbol{q}_R - 2\hat{\boldsymbol{q}}) = 2\boldsymbol{T}_f\, \varepsilon\dot{\hat{\boldsymbol{q}}}$. Mass, momentum and Θ are therefore conserved to round-off for $\varepsilon = 0$, and up to $\mathcal{O}(\varepsilon)$ otherwise.

5 Verification

All numbers come from the Python implementation (`run_tests_theta.py`). The Julia port implements the same algorithm and prints the same tests, with these values in its comments.

5.1 Convergence

Constant frozen coefficients $\bar{\pi} = R$, $\bar{\theta} = 300\,\text{K}$ ($c = 293\,\text{m/s}$), periodic domain of 2000 m, exact normal mode of (4) (pressure amplitude 100 Pa), one acoustic transit.

5.2 Hydrostatic balance

Lapse-rate atmosphere ($T_0 = 288\,\text{K}$, $\Gamma = 6.5\,\text{K/km}$), walls, $[0, 10]\,\text{km}$, $N_e = 40$, $p = 3$, discrete hydrostatic state from Proposition 1 with Θ_h = interpolant of $\bar{\Theta}$.

5.3 Slow forcing

For a random state $\boldsymbol{q}^*$ (background plus noise) and $\boldsymbol{F}_{\text{slow}} = -\mathcal{L}(\boldsymbol{q}^*)$, the state stays at $\boldsymbol{q}^*$: $\max|\boldsymbol{q} - \boldsymbol{q}^*| = 2.8 \cdot 10^{-12}$ (DIRK) and $2 \cdot 10^{-17}$ (IMEX) after 50 steps.

Table 1: L^2 error in Θ; the rates for ρ, m, Θ coincide.

method	p	$N_e = 8$	16	32	64	rate
DIRK, $\varepsilon = 0$	1	6.44e-1	1.30e-1	3.00e-2	7.31e-3	2.04
	2	2.93e-2	3.65e-3	4.56e-4	5.70e-5	3.00
	3	1.39e-3	8.81e-5	5.51e-6	3.44e-7	4.00
DIRK, $\varepsilon = 10^{-9}h/c$	3	1.39e-3	8.81e-5	5.51e-6	3.79e-7	3.86
IMEX, $\varepsilon = 10^{-9}h/c$	1	6.48e-1	1.31e-1	3.00e-2	7.33e-3	2.04
	2	3.20e-2	4.85e-3	9.19e-4	2.07e-4	2.15

Table 2: L^2 error in Θ versus $\varepsilon c/h$ ($N_e = 16$, $p = 3$); the $\varepsilon = 0$ reference is $8.81 \cdot 10^{-5}$ (DIRK) and $1.63 \cdot 10^{-3}$ (IMEX, time-error dominated).

$\varepsilon c/h$	10^{-8}	10^{-7}	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}
DIRK	8.81e-5	8.95e-5	1.79e-4	1.56e-3	1.55e-2	1.55e-1	1.52e+0
IMEX	1.63e-3	1.62e-3	1.48e-3	1.31e-4	1.39e-2	1.53e-1	1.52e+0

5.4 Acoustic pulse between walls

Discrete hydrostatic background plus a 100 Pa pressure pulse ($\Theta' = p'/a$) at 3 km, $N_e = 80$, $p = 3$, $t = 25$ s. The totals are of order $10^4\,\mathrm{kg\,m^{-2}}$ (mass) and $3 \cdot 10^6$ (Θ), so the $\varepsilon = 0$ drifts are at round-off level.

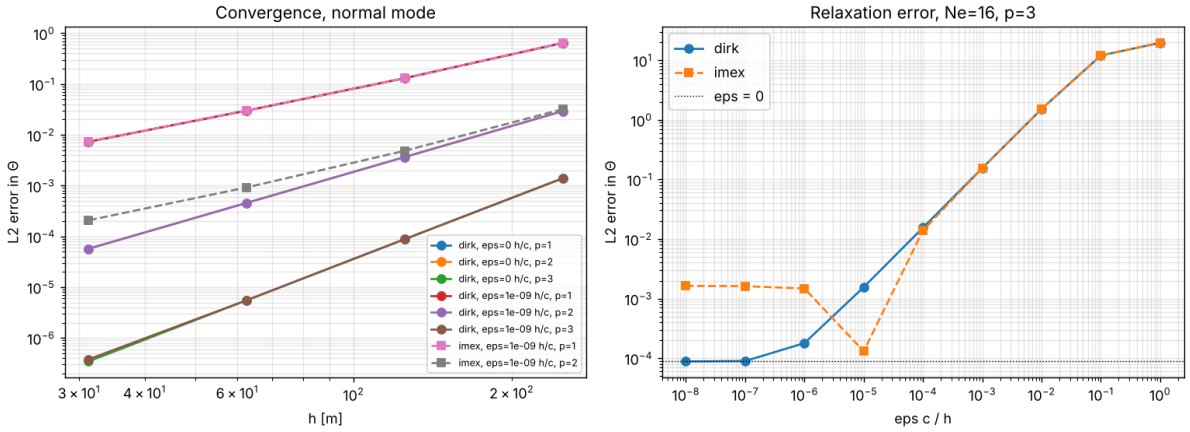


Figure 1: Left: convergence. Right: relaxation error versus $\varepsilon c/h$.

6 Code

`hdg_relax_theta.py`	Python solver (NumPy/SciPy)
`run_tests_theta.py`	Python tests and plots
`julia/HDGRelaxTheta.jl`	Julia module (standard library only)
`julia/run_tests_theta.jl`	Julia tests, writes CSV files

Usage (Julia)

```
include("HDGRelaxTheta.jl"); using .HDGRelaxTheta
atm = Atmosphere(T0 = 288.0, Γ= 0.0065)
```

Table 3: Maximum relative deviation after 200 steps, and residual of a projected background.

	DIRK, $\varepsilon = 0$	DIRK, $\varepsilon = 10^{-3}$	IMEX, $\varepsilon = 10^{-3}$
`pfac`$= 1$	3.2e-13	4.3e-13	5.4e-15
`pfac`$= \gamma$	7.2e-13	8.0e-13	3.8e-15
projected analytic background, $p = 3$, $N_e = 10, 20, 40, 80$: $\max \lvert\dot{\boldsymbol{q}}\rvert$ = 3.1e-6, 3.8e-7, 4.8e-8, 6.1e-9 (rate 3)			

Table 4: Sound speed at $z = 0$, mass and Θ drift, trace residual.

case	$c(0)$ [m/s]	$\lvert\Delta\text{mass}\rvert$	$\lvert\Delta\Theta\rvert$	trace res.
`pfac`$= 1$, $\varepsilon = 0$, DIRK	287.5	2.2e-11	2.2e-8	6.7e-10
`pfac`$= 1$, $\varepsilon = 10^{-4}h/c$, IMEX	287.5	1.4e-5	2.1e-3	1.5e-5
`pfac`$= \gamma$, $\varepsilon = 0$, DIRK	340.2	2.2e-11	5.6e-9	3.3e-10

```
fr = frozen_from_atmosphere(atm; pfac = 1.0) # or FrozenState(pi_fun, theta_fun; ...)
s = HDGRelaxTh(80, 3, 0.0, 1.0e4, fr; bc = :wall, eps = 0.0)
q = balanced_state(s, z -> atm_Theta(atm, z)) # discrete hydrostatic state
F = zeros(s.nq) # slow tendency (split-explicit)
U = run!(s, initial_state(s, q), 0.05, 100; method = :imex, F_slow = F)
z, Q = evaluate(s, U[1:s.nq]) # Q[1,:]=ρ, Q[2,:]=ρw, Q[3,:]=ρθ
```

Python:

```
s = HDGRelaxTheta(Ne, p, zmin, zmax, FrozenState.from_atmosphere(atm, pfac),
                  bc="wall", eps=0.0)
U = s.run(U0, dt, n, method="imex", F_slow=F)
```

For frozen coefficients from a DG slow state, pass functions that evaluate a continuous reconstruction of $\bar{p}/\bar{\Theta}$ and $\bar{\theta}$.

References

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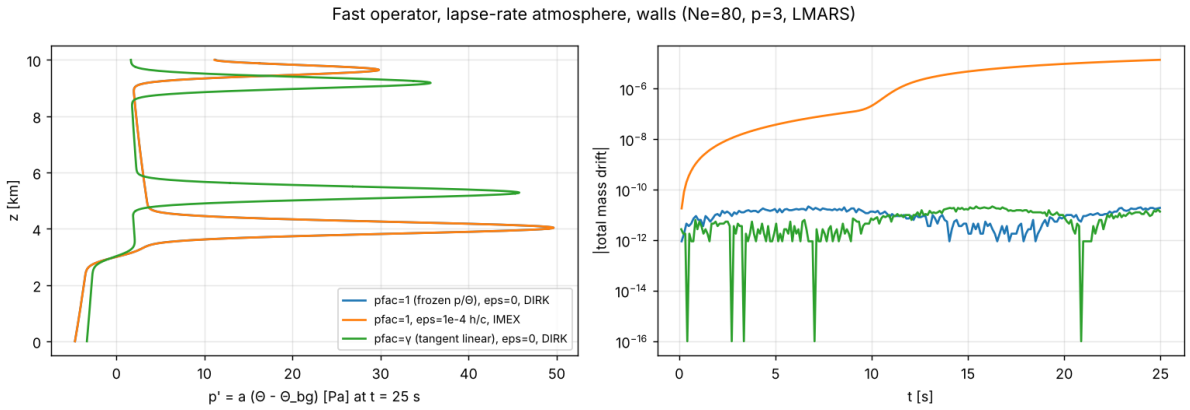


Figure 2: Pressure perturbation $a(\Theta - \Theta_{\mathrm{bg}})$ at $t = 25\,\mathrm{s}$ for `pfac`$= 1$ and γ (different wave speeds), and mass drift.